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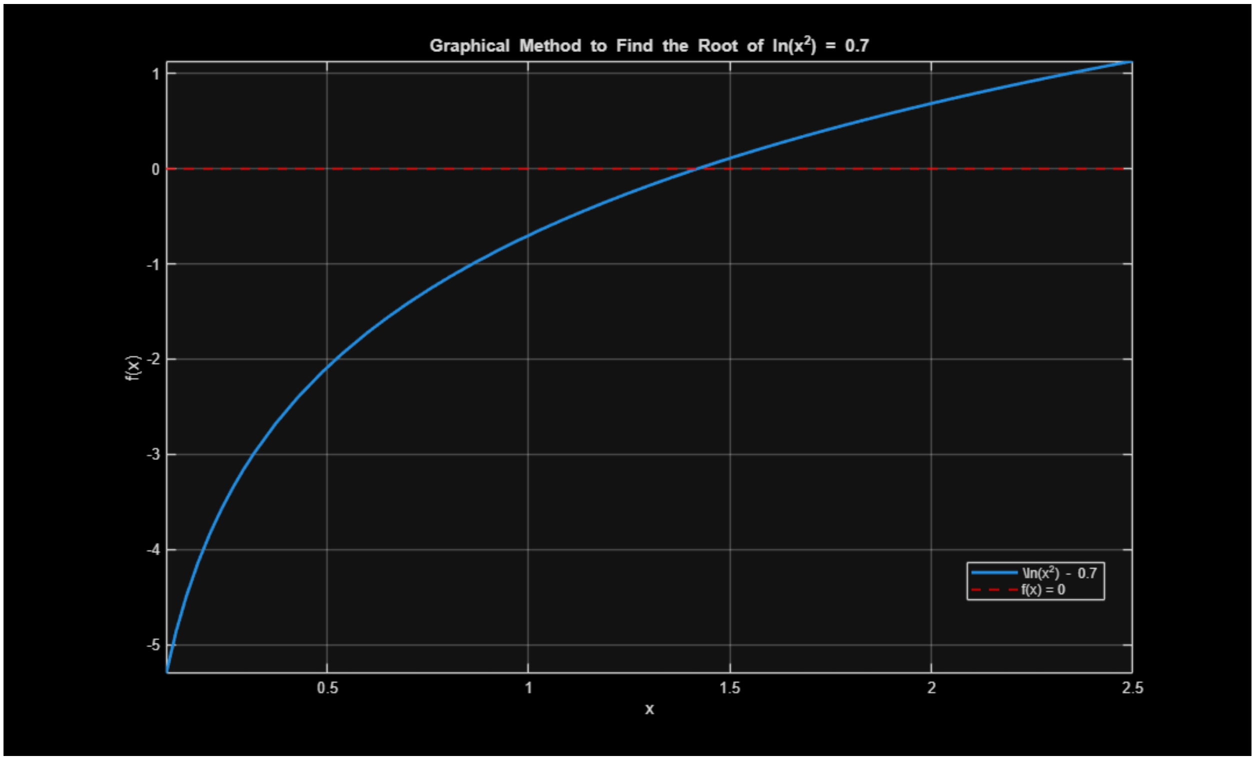
```
clc; clear; close all;
```

Problem #6: Root Finding for ln(x^2) = 0.7

```
f = @(x) log(x.^2) - 0.7;
```

(a) Graphical Method

```
figure;
fplot(f, [0.1, 2.5], 'LineWidth', 2);
grid on;
yline(0, 'r--', 'LineWidth', 1.5); % Reference line for root
xlabel('x');
ylabel('f(x)');
title('Graphical Method to Find the Root of ln(x^2) = 0.7');
legend('\ln(x^2) - 0.7', 'f(x) = 0', 'Location', 'best');
hold on;
```



(b) Bisection Method

```
xL = 0.5;
xU = 2.0;
tol = 1e-5;
max_iter = 100;

fprintf('--- Bisection Method ---\n');
if f(xL) * f(xU) > 0
    error('No root guaranteed in the interval [xL, xU].');
else
    fprintf('%-6s %-10s %-10s %-10s %-10s\n', 'Iter', 'xL', 'xU', 'xR',
'f(xR)');
    for iter = 1:max_iter
        xR = (xL + xU) / 2;
        fxR = f(xR);

        fprintf('%-6d %-10.5f %-10.5f %-10.5f %-10.5f\n', iter, xL, xU, xR,
fxR);

        if abs(fxR) < tol || (xU - xL)/2 < tol
            break;
        end

        if f(xL) * fxR < 0
            xU = xR;
        else
            xL = xR;
```

```
        end
    end
    fprintf('Estimated positive root (Bisection): %.5f\n\n', xR);
end
```

```
--- Bisection Method ---
Iter   xL         xU         xR         f(xR)
1      0.50000    2.00000    1.25000    -0.25371
2      1.25000    2.00000    1.62500    0.27102
3      1.25000    1.62500    1.43750    0.02581
4      1.25000    1.43750    1.34375    -0.10907
5      1.34375    1.43750    1.39062    -0.04049
6      1.39062    1.43750    1.41406    -0.00707
7      1.41406    1.43750    1.42578    0.00944
8      1.41406    1.42578    1.41992    0.00120
9      1.41406    1.41992    1.41699    -0.00293
10     1.41699    1.41992    1.41846    -0.00086
11     1.41846    1.41992    1.41919    0.00017
12     1.41846    1.41919    1.41882    -0.00034
13     1.41882    1.41919    1.41901    -0.00009
14     1.41901    1.41919    1.41910    0.00004
15     1.41901    1.41910    1.41905    -0.00002
16     1.41905    1.41910    1.41908    0.00001
17     1.41905    1.41908    1.41906    -0.00001
Estimated positive root (Bisection): 1.41906
```

(c) False-Position Method

```
xL = 0.5;
xU = 2.0;

fprintf('--- False-Position Method ---\n');
if f(xL) * f(xU) > 0
    error('No root guaranteed in the interval [xL, xU].');
else
    fprintf('%-6s %-10s %-10s %-10s %-10s\n', 'Iter', 'xL', 'xU', 'xR',
'f(xR)');
    xR_old = xL;
    for iter = 1:max_iter

        xR = xU - (f(xU) * (xL - xU)) / (f(xL) - f(xU));
        fxR = f(xR);

        fprintf('%-6d %-10.5f %-10.5f %-10.5f %-10.5f\n', iter, xL, xU, xR,
fxR);

        % Check convergence using relative approximate error or absolute
function value
        if abs(fxR) < tol || abs(xR - xR_old) < tol
            break;
        end
        xR_old = xR;
```

```
        if f(xL) * fxR < 0
            xU = xR;
        else
            xL = xR;
        end
    end
    fprintf('Estimated positive root (False-Position): %.5f\n', xR);
end
```

```
--- False-Position Method ---
Iter    xL          xU          xR          f(xR)
1       0.50000     2.00000     1.62871     0.27557
2       0.50000     1.62871     1.49701     0.10695
3       0.50000     1.49701     1.44840     0.04092
4       0.50000     1.44840     1.43016     0.01557
5       0.50000     1.43016     1.42327     0.00591
6       0.50000     1.42327     1.42066     0.00224
7       0.50000     1.42066     1.41967     0.00085
8       0.50000     1.41967     1.41930     0.00032
9       0.50000     1.41930     1.41915     0.00012
10      0.50000     1.41915     1.41910     0.00005
11      0.50000     1.41910     1.41908     0.00002
12      0.50000     1.41908     1.41907     0.00001
Estimated positive root (False-Position): 1.41907
```

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